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Editor-in-Chief

IEEE Transactions on Power Delivery

Dear Editor-in-Chief,

We submit for your consideration the original research article:

**“Bayesian Three-Hypothesis SPRT for IBR-Aware Transformer
Differential Protection: Waveform Asymmetry, Fault-Integrity Gating,
and 30,000-Trial Monte Carlo Certification”**

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Motivation and problem addressed. Transformer differential (87T) protection has relied on second-harmonic (2H) restraint to distinguish magnetising inrush from internal faults since the 1950s. Inverter-based resources (IBRs) systematically destroy the assumption underlying this restraint: the IBR current controller produces a sinusoidal fault current with $I_2/I_1 < 2\%$, indistinguishable from inrush under the 2H criterion. Concurrently, residual-flux energisation ($B_r \geq 0.7$ T) reduces the inrush second-harmonic ratio to 8–10 %, below the blocking threshold, creating a risk of false trips. Monte Carlo evidence confirms that at $k_{ibr} \geq 0.70$, the conventional relay achieves a detection probability of only $P_D = 0.236$ — one internal fault in four goes undetected. No published method simultaneously addresses all three discriminands (IBR internal fault, magnetising inrush, and CT-saturation-masked external fault) within a single statistical framework with formal error-probability bounds.

Contributions. This paper proposes and validates a three-contribution IBR-aware 87T relay architecture:

1. **Three-hypothesis SPRT on waveform asymmetry (C6):** The per-half-cycle asymmetry index A_k is derived from first principles and shown to be statistically separable across the three hypotheses — inrush ($H_0, \mu_{A_k} = 0.62$), IBR fault ($H_1, \mu_{A_k} = 0.00$), and CT saturation ($H_2, \mu_{A_k} = -0.65$) — with KL divergences of 1.15 and 1.42 nats respectively. Two parallel log-likelihood-ratio accumulators with Wald boundaries achieve $P_D = 1.000$ and median trip time $t_{50} = 20$ ms across 30 000 Monte Carlo trials.
2. **Five-priority integration architecture (C7):** The SPRT is embedded into existing 87T relay logic through five sequentially evaluated priorities, with the SPRT trip signal gated by the fault-integrity index $f_{int} \geq 0.55$ at Priority 4. This restores P_D^{ibr} from 0.236 to 1.000 across 20 000 MC trials and reduces the IBR-fault median trip time from ≥ 60 ms to 20 ms, without hardware modification.
3. **Fault-integrity index with formal separation proof (C8):** The scalar index $f_{int} = 1 - \|\mathbf{r}_n\|/I_{op}$, derived from the LM estimator residual, provides a formally proved 0.34 pu separation between the maximum inrush value ($f_{int}^{H_0} \leq 0.21$) and the minimum internal-fault value ($f_{int}^{H_1} \geq 0.55$), confirmed numerically across all 5 000 Group B (inrush) MC trials with zero overlap.

Validation. The chain is validated on 48 deterministic scenarios across three staged studies (TR-57a analytical, TR-57b EMT-surrogate, TR-64 five-priority integration) with zero misclassification, and on 30 000 MC trials (10 000 Group A IBR fault, 5 000 Group B inrush, 5 000 Group C CT-saturation, 10 000 Group D SG fault). All groups achieve rate = 1.000 with 95 % Wilson CI lower bound ≥ 0.9992 , meeting IEC 60255-111 performance targets.

Fit for *IEEE Transactions on Power Delivery*. The paper addresses the transformer protection reliability gap created by IBR integration — a problem directly relevant to the Transactions’ readership of protection

engineers, utility planners, and relay manufacturers. The combination of Wald's sequential hypothesis testing, formal distribution derivation from physical models (Jiles–Atherton, Park, CT knee-point saturation), and 30 000-trial MC certification delivers the analytical and empirical depth appropriate for a Transactions paper. The work is original and has not been submitted elsewhere.

Suggested reviewers.

- Prof. Ali Hooshyar, University of Toronto (IBR-impacted protection, transformer differential relaying)
- Prof. Bogdan Kasztenny, SEL Engineering (transformer inrush discrimination, differential relay design)
- Prof. Massimo Mitolo, Irvine Valley College / IEEE PSRC (power transformer protection, harmonic analysis)
- Dr. Evangelos Farantatos, EPRI (IBR fault characterisation and protection impacts)

Authorship and conflicts. Both authors declare no conflicts of interest. The contributions statement follows separately. Simulation code and parametric data are available upon request.

Yours sincerely,

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